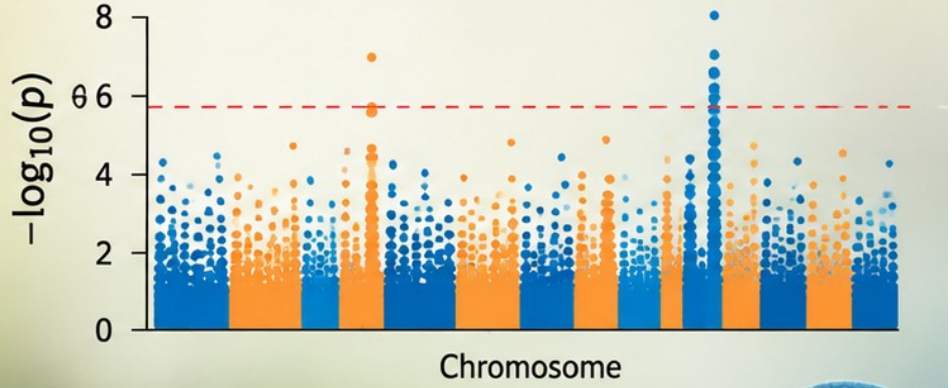
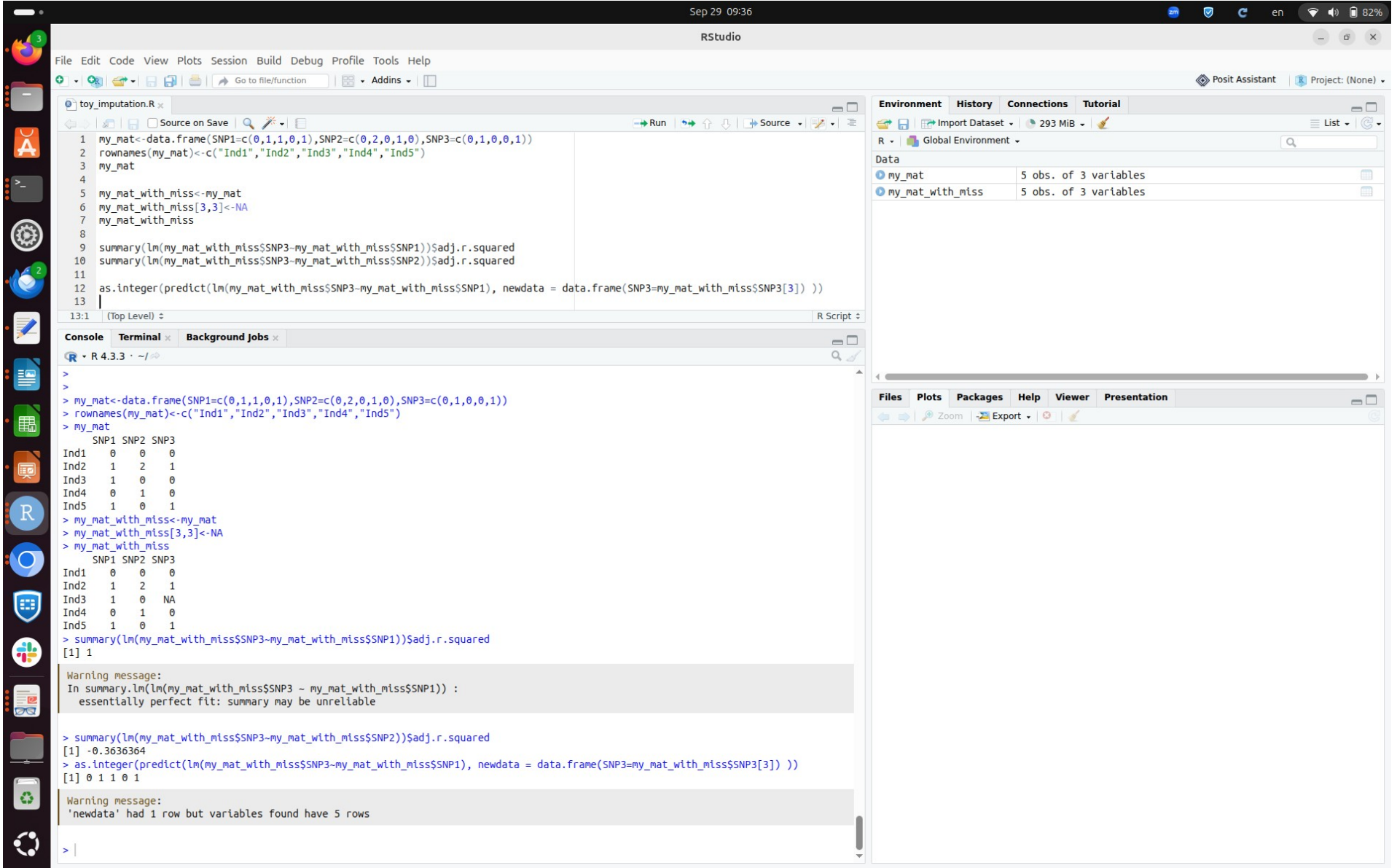
Genome-Wide Association Studies (GWAS) for Evolutionary Biology

Nikolay Oskolkov, Group Leader (PI) at NIRI, Riga, Latvia

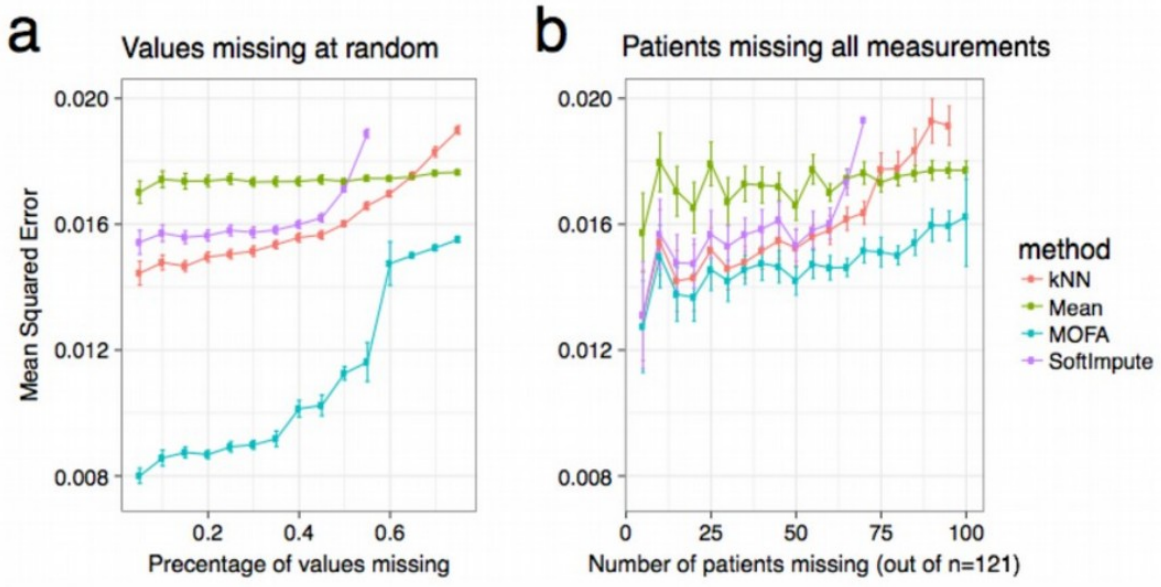


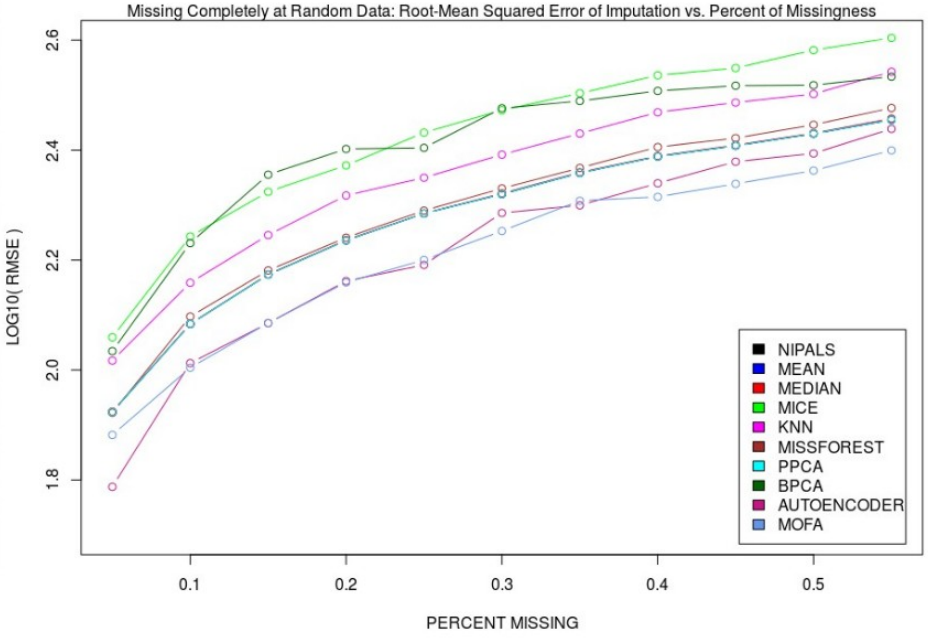


Missing data and general imputation principles


Sep 29 09:36
en
82%
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Posit Assistant
Project: (None)
toy_imputation.R
Source on Save
Run
Source
1 my_mat<-data.frame(SNP1=c(0,1,1,0,1),SNP2=c(0,2,0,1,0),SNP3=c(0,1,0,0,1))
2 rownames(my_mat)<-c("Ind1","Ind2","Ind3","Ind4","Ind5")
3 my_mat
4
5 my_mat_with_miss<-my_mat
6 my_mat_with_miss[3,3]<-NA
7 my_mat_with_miss
8
9 summary(lm(my_mat_with_miss$SNP3~my_mat_with_miss$SNP1))$adj.r.squared
10 summary(lm(my_mat_with_miss$SNP3~my_mat_with_miss$SNP2))$adj.r.squared
11
12 as.integer(predict(lm(my_mat_with_miss$SNP3~my_mat_with_miss$SNP1), newdata = data.frame(SNP3=my_mat_with_miss$SNP3[3])))
13
13:1 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.3.3 · ~/
> my_mat<-data.frame(SNP1=c(0,1,1,0,1),SNP2=c(0,2,0,1,0),SNP3=c(0,1,0,0,1))
> rownames(my_mat)<-c("Ind1","Ind2","Ind3","Ind4","Ind5")
> my_mat
SNP1 SNP2 SNP3
Ind1 0 0 0
Ind2 1 2 1
Ind3 1 0 0
Ind4 0 1 0
Ind5 1 0 1
> my_mat_with_miss<-my_mat
> my_mat_with_miss[3,3]<-NA
> my_mat_with_miss
SNP1 SNP2 SNP3
Ind1 0 0 0
Ind2 1 2 1
Ind3 1 0 NA
Ind4 0 1 0
Ind5 1 0 1
> summary(lm(my_mat_with_miss$SNP3~my_mat_with_miss$SNP1))$adj.r.squared
[1] 1
Warning message:
In summary.lm(lm(my_mat_with_miss$SNP3 ~ my_mat_with_miss$SNP1)) :
essentially perfect fit: summary may be unreliable
> summary(lm(my_mat_with_miss$SNP3~my_mat_with_miss$SNP2))$adj.r.squared
[1] -0.3636364
> as.integer(predict(lm(my_mat_with_miss$SNP3~my_mat_with_miss$SNP1), newdata = data.frame(SNP3=my_mat_with_miss$SNP3[3])))
[1] 0 1 1 0 1
Warning message:
'newdata' had 1 row but variables found have 5 rows
>
Environment
History
Connections
Tutorial
Import Dataset
293 MiB
List
R
Global Environment
Data
my_mat 5 obs. of 3 variables
my_mat_with_miss 5 obs. of 3 variables
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export

MOFA: imputation of missing values

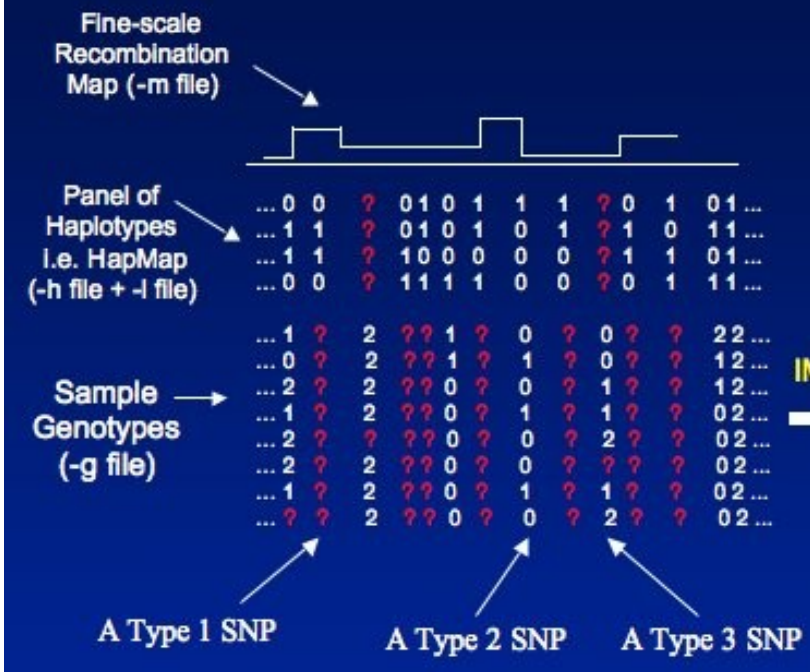




Bayesian framework is insensitive to missing data, priors compensate for the lack of data

Genetic imputation and phasing

Overview of Imputation using IMPUTE



Type 1 SNP : in haplotypes file only

Type 2 SNP : in haplotypes and genotypes file

Type 3 SNP : in genotypes file only

IMPUTATION

Imputed Genotypes (-o file + -i file)

The above figure over-simplifies what IMPUTE does. The output for each genotype is a probability distribution on genotypes i.e.

0	1	2
0.01	0.18	0.81

This captures the uncertainty in the prediction.

IMPUTE2

- Home
- Getting Started
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- Whole Chromosomes
- Merging Panels
- Concordance Tables
- Scripts
- FAQ
- References
- Contributors
- Mail List

Last update: **23** Dec 2014

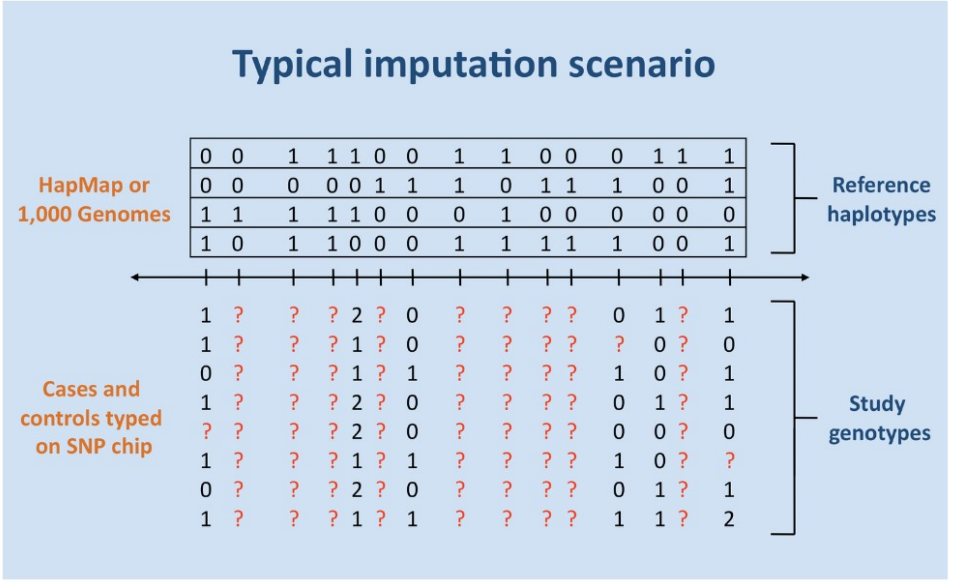
IMPUTE2

IMPUTE version 2 (also known as **IMPUTE2**) is a genotype imputation and haplotype phasing program based on ideas from Howie *et al.* 2009:

B. N. Howie, P. Donnelly, and J. Marchini (2009) **A flexible and accurate genotype imputation method for the next generation of genome-wide association studies**. *PLoS Genetics* 5(6): e1000529 [Open Access Article] [Supplementary Material]

IMPUTE2 also includes features that were introduced in other publications, which you can find here.

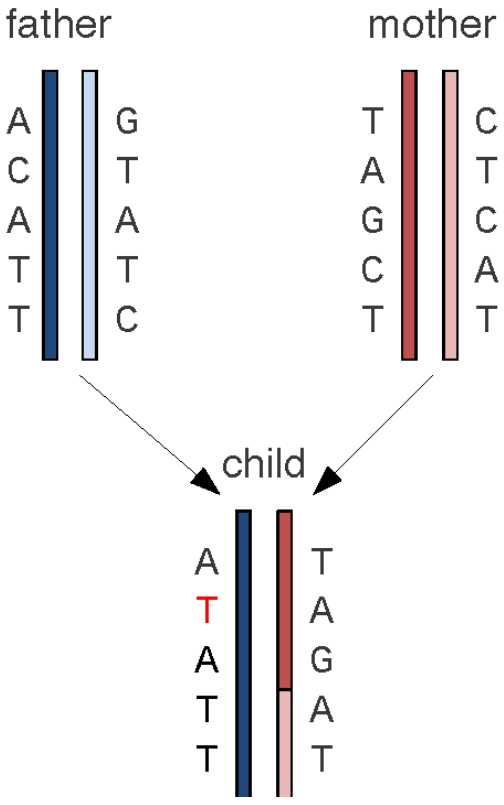
The figure below shows the most common scenario in which imputation is used: unobserved genotypes (red question marks) in a set of study individuals are imputed (or predicted) using a set of reference haplotypes and genotypes from a SNP chip.



Phasing

Genotyping tells us which alleles an individual carries at each position.

Phasing tells us which alleles are located together on the same chromosome.



SHAPEIT

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Download and Licence
Contact

Getting started
Algorithm parameters
Model parameters
Input file options
Output file options

Pre-phasing Imputation
Using family information
Using a reference panel
Phasing X chromosomes
Read aware phasing
Using phasing uncertainty
Genotype calling

Options list
File format specification
Version history

Last update: **17** January 2018

SHAPEIT

SHAPEIT is a fast and accurate method for estimation of haplotypes (aka phasing) from genotype or sequencing data.

SHAPEIT has primarily been developed by Dr Olivier Delaneau through a collaborative project between the research groups of Prof Jean-Francois Zagury at CNAM and Prof Jonathan Marchini at Oxford. Funding for this project has been received from several sources : CNAM, Peptinov, MRC, Leverhulme, The Wellcome Trust.

SHAPEIT has several notable features:

- Linear complexity with the number of SNPs and conditioning haplotypes.
- Whole chromosome GWAS scale datasets can be phased in a single run.
- Phasing individuals with any level of relatedness
- Phasing is multi-threaded to tailor computational times to your resources.
- Handles X chromosomes phasing.
- Phasing using a reference panel (eg.1,000 Genomes) to aid phasing
- Ideal for pre-phasing imputation together with IMPUTE2

SHAPEIT is free for academic use only. For commercial use please see here

SHAPIT2: Hidden Markov Model (HMM)

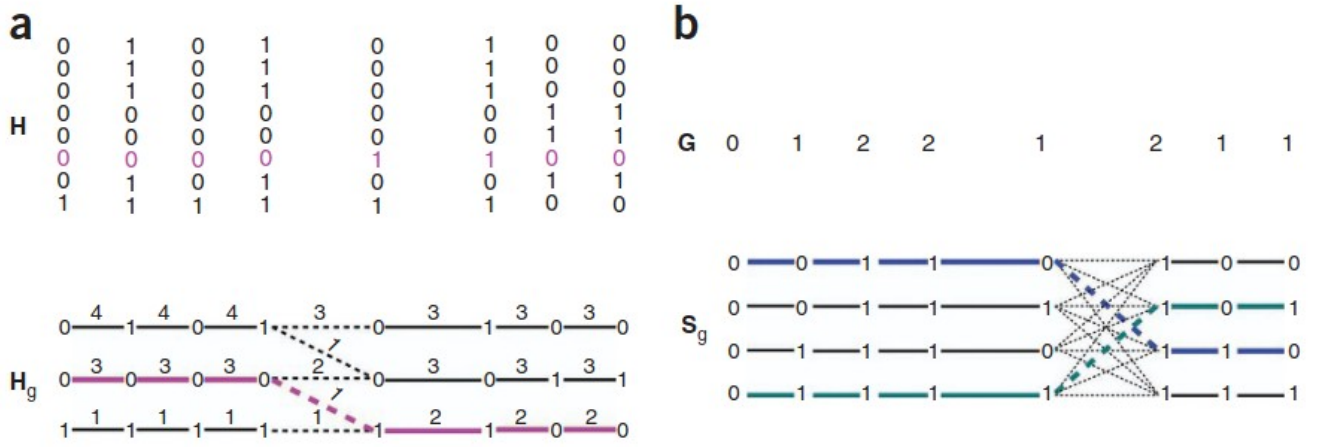


Figure 1 | Illustration of the model and the associated graphs for a simple example. (**a**,**b**) In this example, **H** contains $K = 8$ haplotypes (rows in **a**) and the individual's genotype **G** contains four heterozygous SNPs (**b**), both defined over $M = 8$ markers (columns). In **a**, we illustrate how the graph $\mathbf{H}_g$ is built by splitting the haplotypes of **H** between markers 4 and 5, resulting in two segments that each contain $J = 3$ distinct haplotypes. The nodes of the graph are labeled either with allele 1 or allele 0. Each edge is weighted by the number of haplotypes in **H** that traverse it. A haplotype of **H** and its corresponding path in $\mathbf{H}_g$ is illustrated in magenta. In **b**, we illustrate how the graph $\mathbf{S}_g$ is built by making two segments of five and three SNP markers, each one containing two heterozygous markers in **G** (represented as state 1; state 0 and 2 are wild type and homozygous, respectively). Each segment has four possible haplotypes compatible with **G**. A pair of paths in $\mathbf{S}_g$ compatible with **G** is colored blue and green.

faculty.washington.edu/browning/beagle/beagle.html

Beagle 5.5

Program: beagle 5.5 (version: 27Feb25.75f)
Author: Brian Browning
Email: browning@uw.edu

Contents

- Introduction
- Citation
- Download Beagle 5.5

Introduction

Beagle is a software package for phasing genotypes and imputing ungenotyped markers. Beagle version 5.5 has improved memory efficiency and accuracy when analyzing large sequence data sets.

[top]

Citation

If you use Beagle in a published analysis, please report the program version and cite the appropriate article.

The Beagle 5.5 genotype phasing method is described in:

> B L Browning, X Tian, Y Zhou, and S R Browning (2021) Fast two-stage phasing of large-scale sequence data. Am J Hum Genet 108(10):1880-1890. doi:10.1016/j.ajhg.2021.08.005

The Beagle 5.5 genotype imputation method is described in:

> B L Browning, Y Zhou, and S R Browning (2018). A one-penny imputed genome from next generation reference panels. Am J Hum Genet 103(3):338-348. doi:10.1016/j.ajhg.2018.07.015

[top]

Download Beagle 5.5

Beagle is free software: you can redistribute it and/or modify it under the terms of the GNU General Public License as published by the Free Software Foundation, either version 3 of the License, or (at your option) any later version.git

ARTICLE

A One-Penny Imputed Genome from Next-Generation Reference Panels

Brian L. Browning,[1,2,*] Ying Zhou,[2] and Sharon R. Browning[2]

Genotype imputation is commonly performed in genome-wide association studies because it greatly increases the number of markers that can be tested for association with a trait. In general, one should perform genotype imputation using the largest reference panel that is available because the number of accurately imputed variants increases with reference panel size. However, one impediment to using larger reference panels is the increased computational cost of imputation. We present a new genotype imputation method, Beagle 5.0, which greatly reduces the computational cost of imputation from large reference panels. We compare Beagle 5.0 with Beagle 4.1, Impute4, Minimac3, and Minimac4 using 1000 Genomes Project data, Haplotype Reference Consortium data, and simulated data for 10k, 100k, 1M, and 10M reference samples. All methods produce nearly identical accuracy, but Beagle 5.0 has the lowest computation time and the best scaling of computation time with increasing reference panel size. For 10k, 100k, 1M, and 10M reference samples and 1,000 phased target samples, Beagle 5.0's computation time is 3× (10k), 12× (100k), 43× (1M), and 533× (10M) faster than the fastest alternative method. Cost data from the Amazon Elastic Compute Cloud show that Beagle 5.0 can perform genome-wide imputation from 10M reference samples into 1,000 phased target samples at a cost of less than one US cent per sample.

Introduction

Genotype imputation uses a reference panel of phased, sequenced individuals to estimate sequence data in target samples that have been genotyped on a SNP array.[1,2] Imputation of ungenotyped markers is a standard tool in genome-wide association studies because it greatly increases the number of markers that can be tested for association with a trait. Genotype imputation also provides a foundation for meta-analysis of genome-wide association studies because it converts data for samples that have been genotyped on different SNP arrays into genotype data for a shared set of sequence variants. Meta-analyses based on genotype imputation have discovered thousands of new genetic associations.[3]

The reference panel plays the primary role in determining the accuracy of imputed variants. Imputation accuracy for a variant generally increases with increasing reference panel size, and variants must be present in the reference panel in order to be accurately imputed.[4,5] Consequently, whenever a significantly larger reference panel becomes available, it is advantageous to re-impute target samples with the larger panel for subsequent analysis.

The value of larger reference panels has led to steadily increasing reference panel size, with the largest reference panels to date having tens of thousands of samples.[6–12] Ongoing, large-scale projects, such as the Trans-Omics for Precision Medicine (TopMed) program[13] and the National Human Genome Research Institute's Centers for Common

Increasing reference panel size increases the computational cost of imputation, and this increasing cost has motivated the development of many new computational methods and optimizations. Current imputation methods are able to make use of a rich palette of computational techniques, including the use of identity-by-descent,[6,14,15] haplotype clustering,[4,16] and linear interpolation[5] to reduce the model state space, the use of pre-phasing to reduce computational complexity,[14,17,18] and the use of specialized reference file formats to reduce file size and memory footprint.[4,5,19] The techniques developed thus far have made it possible to provide imputation using reference panels with tens of thousands of individuals as a free web service.[4]

However, the total computational cost of imputation is substantial and increasing. Millions of individuals have been genotyped on SNP arrays and it is probable that millions more will be genotyped in the future. A sample may be re-imputed to a current state-of-the-art reference panel each time it is analyzed or included in a meta-analysis. The increasing size of reference panels increases the per-sample cost of re-imputation. If the past exponential growth in reference panel size continues, reference panels could have ten million individuals within a decade.[6,7,10–12]

In this paper, we present new methodology that reduces the computational cost of genotype imputation 500-fold when imputing from 10M reference samples, and we use the Amazon Elastic Compute Cloud to show that the new methodology can impute genotypes from 10M reference samples into 1,000 phased target samples for a cost of less than one US penny per sample.

BEAGLE reference dataset at genomic positions that were not found in the original 1000 Genomes dataset. Upon further investigation, most of these could be attributed to one-off position differences with some indels reported in the 1000 Genomes dataset. This demonstrates how different data processing pipelines handle complex genotype information in slightly different ways. For the same reason, the total number of markers at unique positions differs in each version of the reference dataset.

Results

All programs outperformed others in certain areas. Based on all of the metrics measured, IMPUTE2 seemed to perform with the greatest accuracy and quality although other programs performed better in other areas.

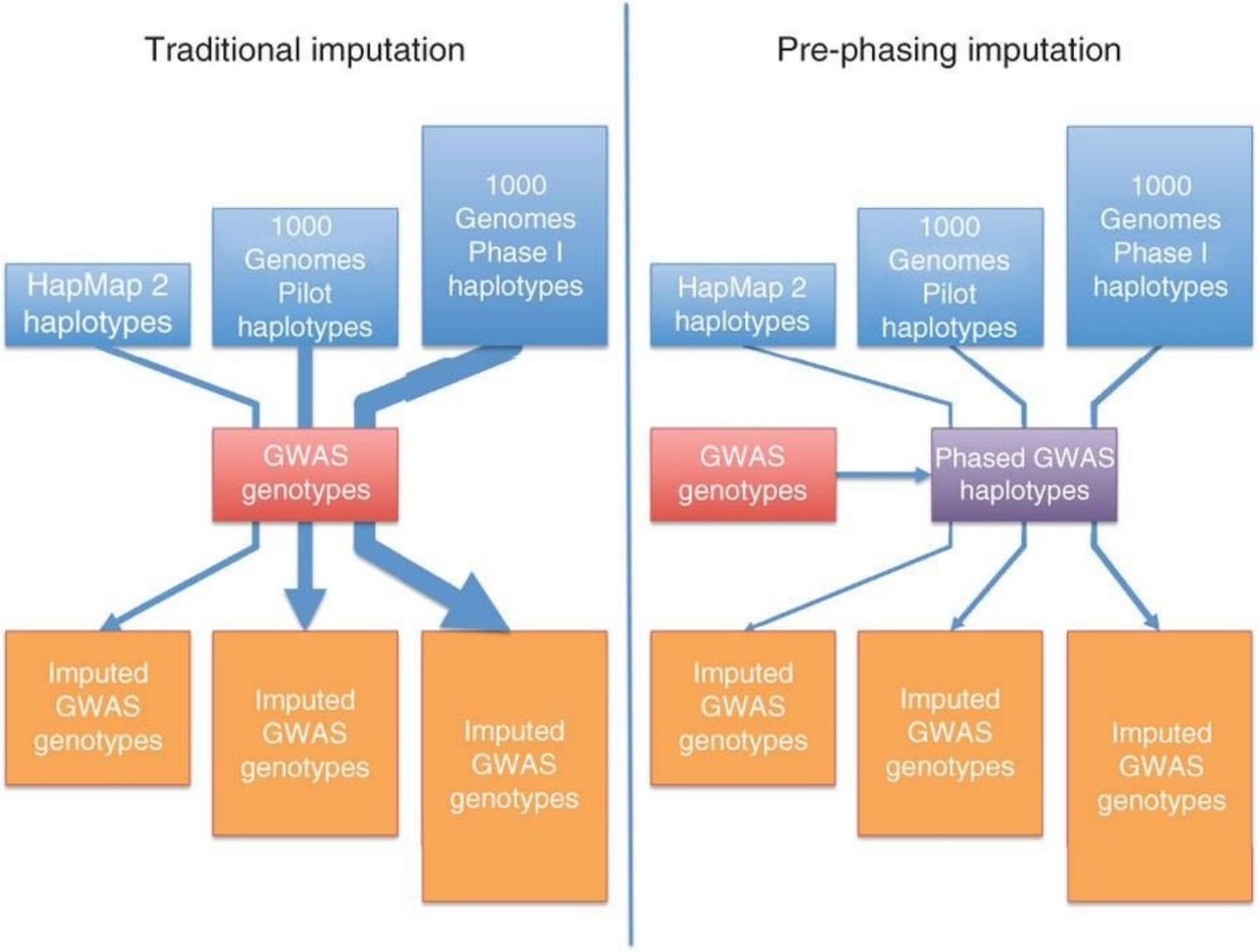
As expected, pre-phasing the original dataset drastically improved the total compute time. When the data was pre-phased, IMPUTE2 ran the quickest, followed by Minimac, and then BEAGLE. Without pre-phasing, IMPUTE2 was much faster than BEAGLE.

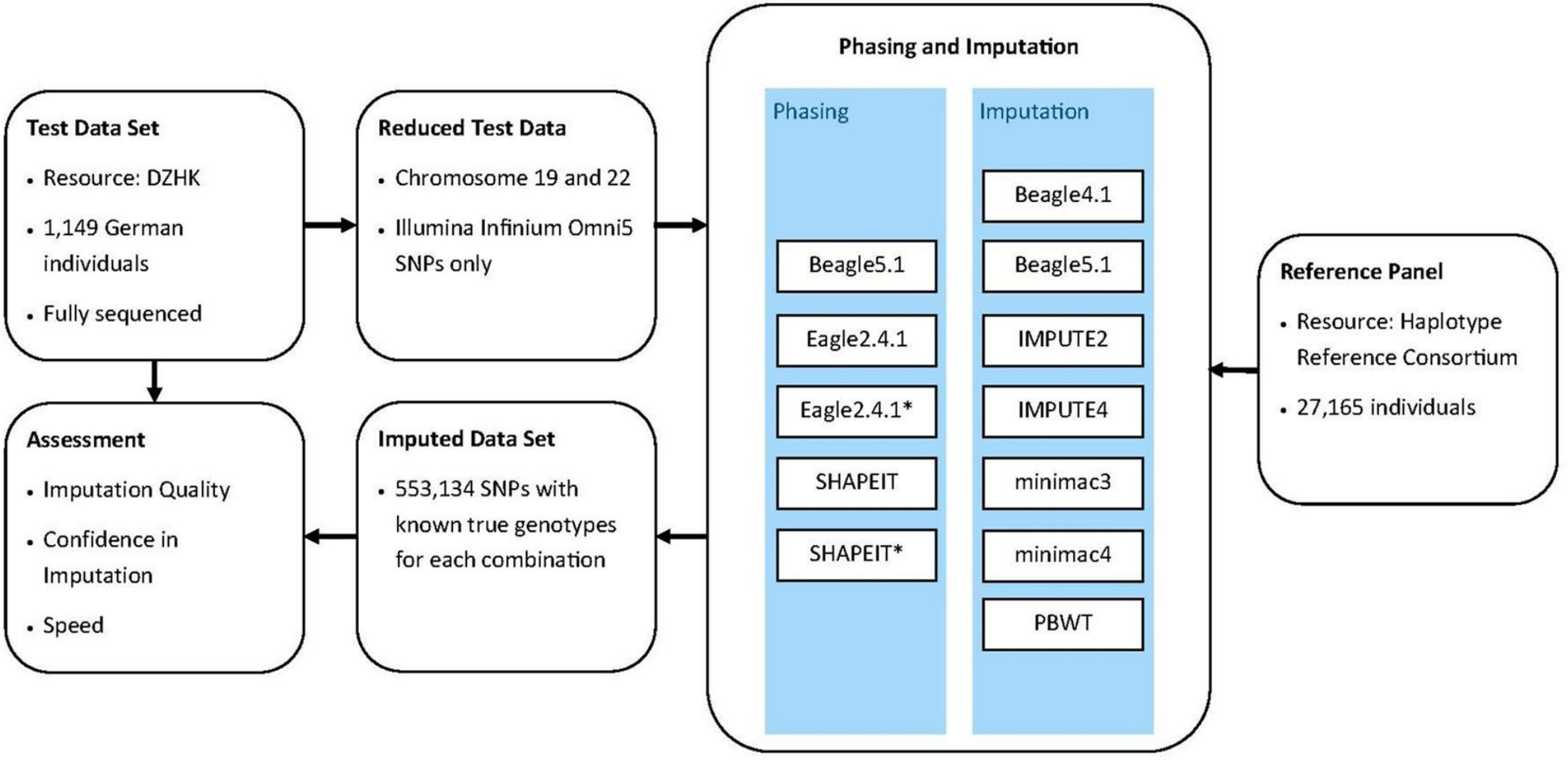
IMPUTE2 also had superior concordance rates, although all software programs performed well in this area. Minimac had the lowest concordance rate at 96.25%.

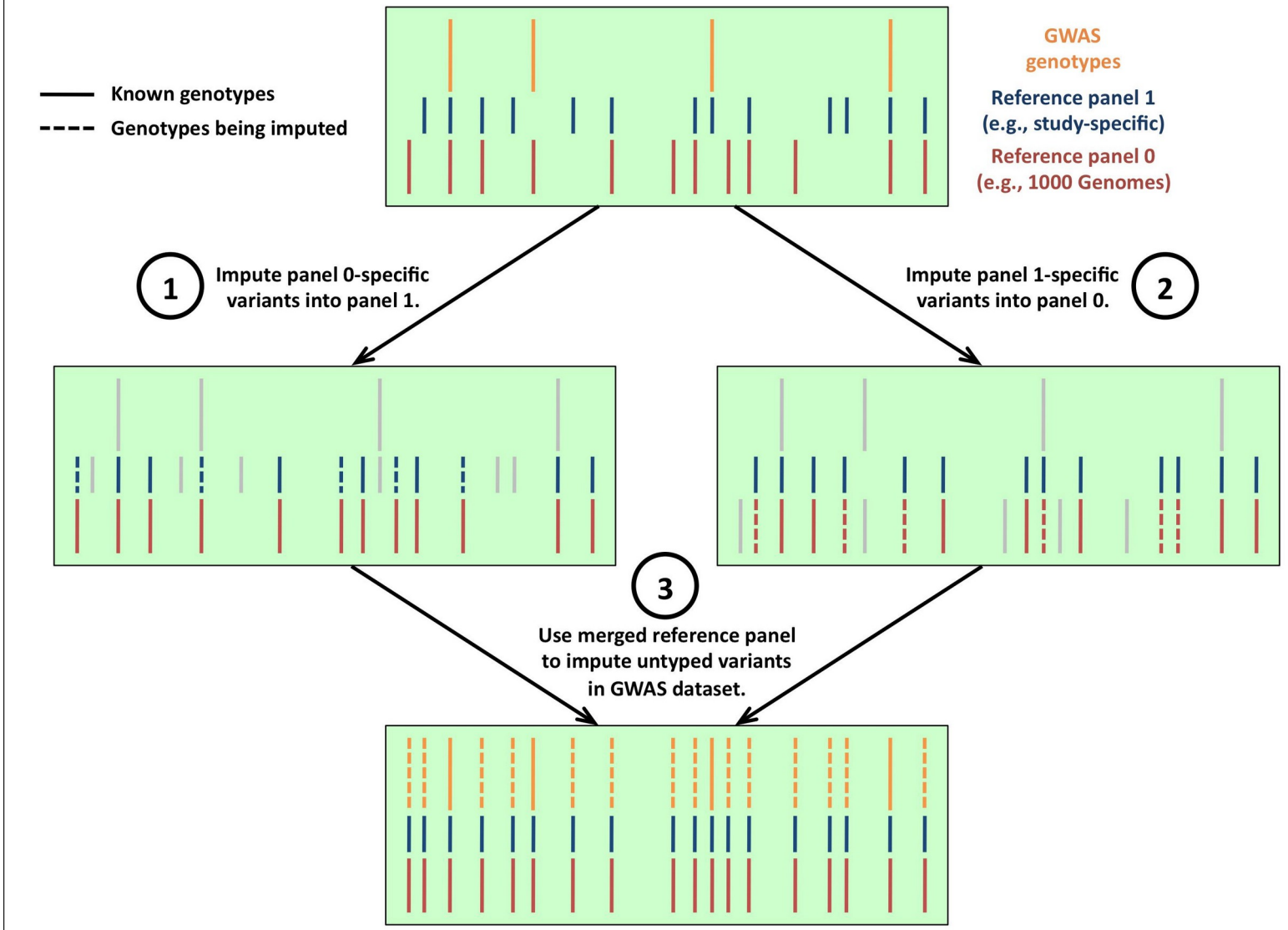
Software	Total Compute Time*	Mean SNP Concordance	Total # SNPs	# High Quality SNPs	% High Quality Imputed
IMPUTE2	23 hours	99.98%	668,180	620,792	92.9%
BEAGLE	213 hours	98.43%	484,023	320,991	66.3%
IMPUTE2 with Pre-phasing	8 hours	99.92%	668,180	297,196	44.5%
BEAGLE with Pre-phasing	34 hours	98.05%	484,023	293,890	60.7%
Minimac	18 hours	96.25%	667,870	450,790	67.5%

**includes all steps required*

Without pre-phasing, IMPUTE2 had the highest quality imputation, but after pre-phasing, the certainty metric provided in the IMPUTE2 output dropped dramatically (see first figure below). The R^2 accuracy value given by BEAGLE was also lower in the output based on pre-phased data, but the change was not nearly as dramatic (see


Traditional imputation
HapMap 2 haplotypes
1000 Genomes Pilot haplotypes
1000 Genomes Phase I haplotypes
GWAS genotypes
Imputed GWAS genotypes
Imputed GWAS genotypes
Imputed GWAS genotypes
Pre-phasing imputation
HapMap 2 haplotypes
1000 Genomes Pilot haplotypes
1000 Genomes Phase I haplotypes
GWAS genotypes
Phased GWAS haplotypes
Imputed GWAS genotypes
Imputed GWAS genotypes
Imputed GWAS genotypes


Test Data Set
Resource: DZHK
1,149 German individuals
Fully sequenced
Reduced Test Data
Chromosome 19 and 22
Illumina Infinium Omni5 SNPs only
Phasing and Imputation
Phasing
Beagle5.1
Eagle2.4.1
Eagle2.4.1*
SHAPEIT
SHAPEIT*
Imputation
Beagle4.1
Beagle5.1
IMPUTE2
IMPUTE4
minimac3
minimac4
PBWT
Reference Panel
Resource: Haplotype Reference Consortium
27,165 individuals
Assessment
Imputation Quality
Confidence in Imputation
Speed
Imputed Data Set
553,134 SNPs with known true genotypes for each combination


Known genotypes
Genotypes being imputed
GWAS genotypes
Reference panel 1 (e.g., study-specific)
Reference panel 0 (e.g., 1000 Genomes)
1
Impute panel 0-specific variants into panel 1.
2
Impute panel 1-specific variants into panel 0.
3
Use merged reference panel to impute untyped variants in GWAS dataset.